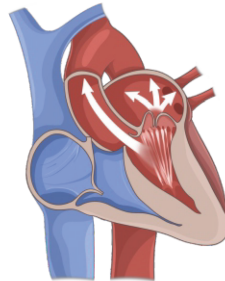




MITRAL VALVE PROLAPSE

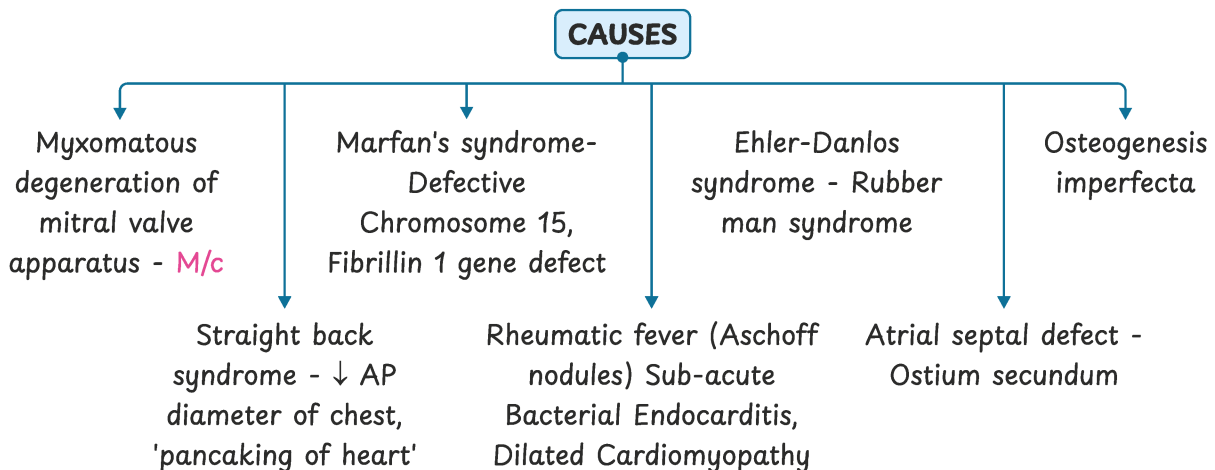
MITRAL VALVE PROLAPSE/ FLOPPY VALVE SYNDROME / BARLOW SYNDROME

- Prolapse occurring towards left atria → undue stretching of loose chordae tendinae → Mid-systolic click



CAUSES

00:01:21

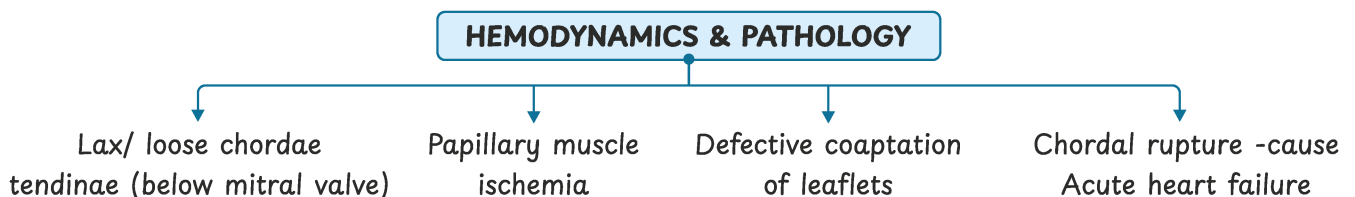


Important Information

- M/c cause of mitral regurgitation - Mitral valve prolapse

HEMODYNAMICS & PATHOLOGY

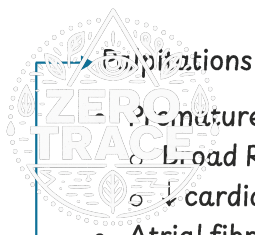
00:07:25



CLINICAL FEATURES

00:09:52

- Severity - Males > Females
- Asymptomatic - M/c



Palpitations

Premature Ventricular Contractions

- Load R & S wave
- ↓ cardiac output → dizziness

- Atrial fibrillation
 - Irregular RR interval, No P waves

Recurrent syncopal attacks

Chest pain at rest

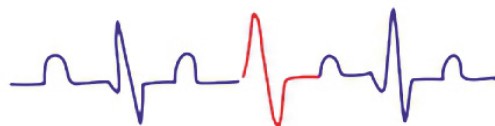
Valve endothelial damage

- Platelet plug formation
 - If embolized - Transient Ischemic attacks
 - T/t - Dual anti-platelet therapy

↑Risk of infective endocarditis, sudden cardiac death

Chordal rupture

- Acute Mitral Regurgitation
- Severe pulmonary edema
- Acute decompensated congestive Heart failure

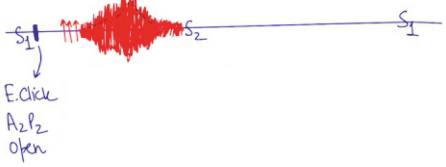
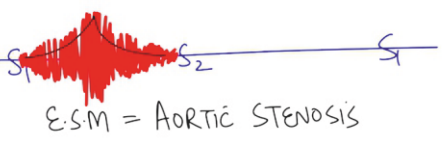
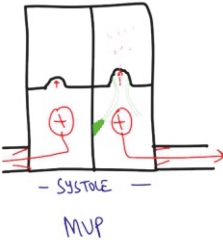
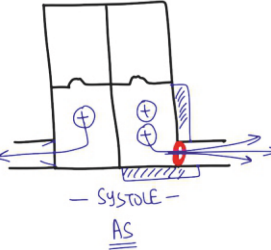


A-Fib



EXAMINATION FINDINGS

00:15:56

MITRAL VALVE PROLAPSE	AORTIC STENOSIS
• Mid-systolic click ◦ Cause - Stretching of chordae tendinae • Late systolic murmur / Late systolic crescendo-decrescendo murmur ◦ Cause - Excess bulge of mitral (& tricuspid) valve → defective coaptation of leaflets (later part of systole) → leakage of blood from left ventricle to left atria → murmur	• Mid systolic / ejection systolic murmur (crescendo- decrescendo murmur) ◦ Cause - Left ventricular hypertrophy ◦ As LV contracts → ↑ ejection velocity of blood (small amount of blood) → more turbulence produced ◦ Associated with carotid thrill
 E.click A2P2 open	 E.S.M = AORTIC STENOSIS
 - SYSTOLE - MVP	 - SYSTOLE - AS



- **MR** involved in Mitral valve prolapse - Posterior leaflet > anterior leaflet

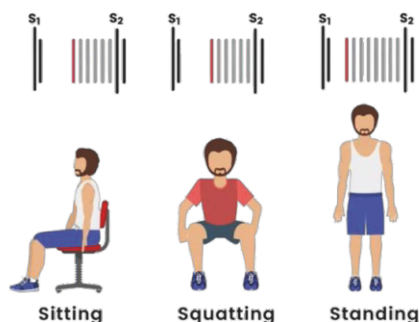
POSTERIOR LEAFLET AFFECTED	ANTERIOR LEAFLET AFFECTED
• Jet of blood moves anteriorly • Murmur radiates to base of heart	• Jet of blood moves posteriorly • Murmur radiates to axilla

Important Information

- **Mitral regurgitation (MR)** - Pan systolic murmur radiating to axilla
- Mitral valve prolapse (MVP) - Late systolic murmur radiating to axilla (anterior leaflet affected)

DYNAMIC AUSCULTATION

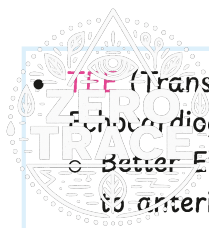
- Prolapse $\propto \frac{1}{\text{Volume of blood in cardiac chamber}}$



On squatting	On standing
On squatting ↓ ↑ Venous return ↓ ↑ Blood volume in cardiac chamber ↓ ↓ Prolapse ↓ Late mid-systolic click ↓ Shorter Murmur duration	On standing ↓ ↓ Venous return ↓ ↓ Blood volume in cardiac chamber ↓ ↑ Prolapse ↓ Early mid-systolic click ↓ Longer Murmur duration

WORK-UP

ECHOCARDIOGRAPHY (IOC)	ECG	T/t
• TTE (Transthoracic Echocardiography) ◦ Bulge > 2 mm above mitral annulus	• T wave inversion ◦ Cause - papillary muscle ischemia	• β blockers - for palpitation • Oral anticoagulants- for Atrial fibrillation



- | | | |
|---|--|--|
| <ul style="list-style-type: none">• TTE (Transesophageal Echocardiography)<ul style="list-style-type: none">○ Better Evaluation of damage to anterior & posterior leaflets | <ul style="list-style-type: none">• Broad QRS complexes (with normal sinus rhythm)<ul style="list-style-type: none">○ Premature Ventricular Contractions• Atrial fibrillation | <ul style="list-style-type: none">• Dual antiplatelet therapy (↓ dose aspirin 75 mg) - for Transient Ischemic Attacks• Mitral valve repair/ replacement (Main T/t)• Prophylactic antibiotics - Previous episode of infective endocarditis |
|---|--|--|

Important Information

- Preferred (for Mitral valve prolapse) - Cardiac MRI > ECHO



- Q. A 28-year-old female came to the Emergency department with complaints of palpitations and dyspnea. On auscultation, mid systolic clicks are heard. All of the following are features of the condition except?
- a. Transient ischemic attacks
 - b. Early systolic murmur
 - c. Myxomatous degeneration of chordae tendineae
 - d. Infective endocarditis

Ans: (b)

- Q. 30-year-old female presents to the clinic due to occasional palpitations, mild exertional shortness of breath and chest discomfort. Her blood pressure is 120/80 mmHg, HR is 82 bpm and regular. There is a mid-systolic click heard best at the apex followed by a late systolic murmur. On squatting the murmur of this condition will show the following impact ?
- a. Shorter in duration
 - b. Louder
 - c. No impact on duration or intensity
 - d. Longer in duration

Ans: (a)

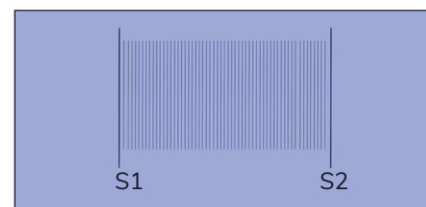
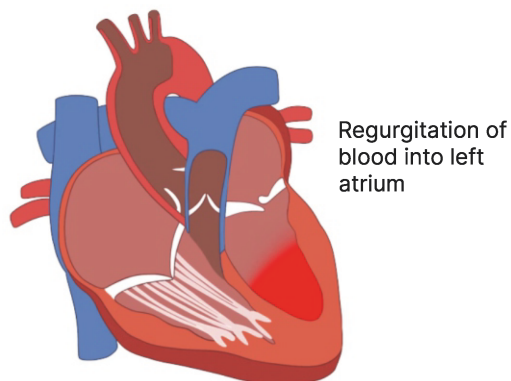
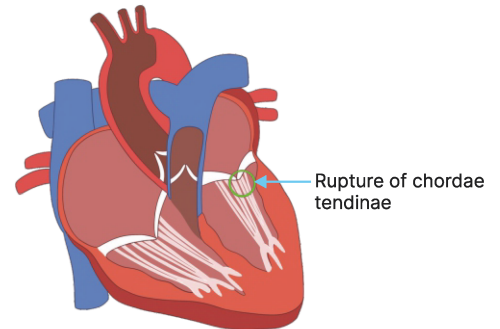


MITRAL REGURGITATION



MITRAL REGURGITATION

- M/c valvular lesion after MI - Acute onset mitral regurgitation
- M/c valvular lesion in children (with rheumatic heart disease) - Mitral regurgitation
- M/c valvular lesion in infective endocarditis - Mitral regurgitation

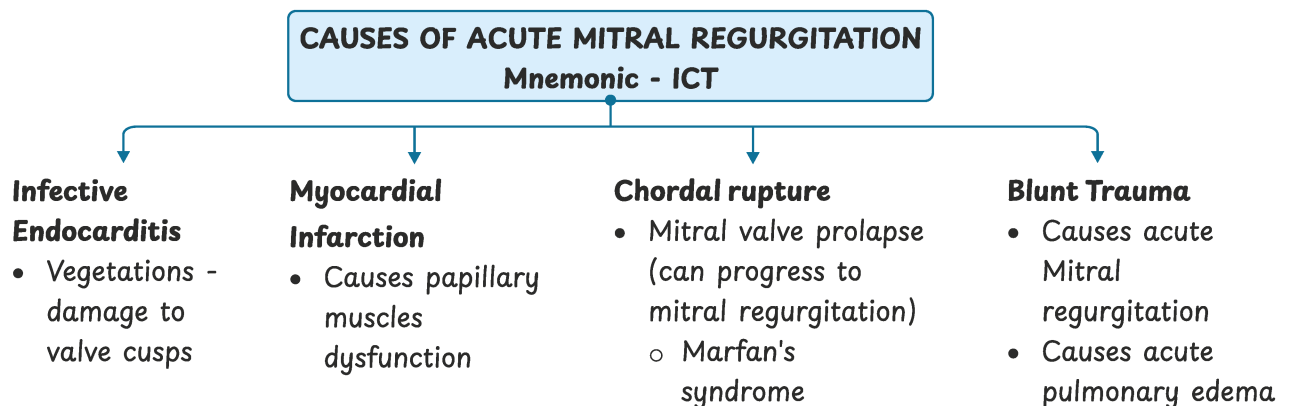


Pan systolic murmur

- Leakage of blood from left ventricle to left atrium (chronic mitral regurgitation)
 - Dysfunction of valve leaflets → Congestion of left atria → leak to pulmonary system → Pulmonary edema & orthopnea
 - Murmur of uniform intensity produced (all throughout systole) → Pan systolic murmur

CAUSES

00:03:39



Important Information

- Causes of sudden death due to Blunt trauma (chest)
 - Commotio cordis



- Trauma received at middle of T-wave (vulnerable period of heart) → progresses to ventricular fibrillation
- Hemopericardium
- Acute onset Mitral regurgitation

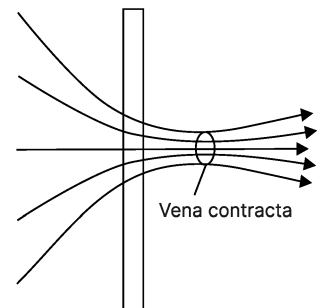
CAUSES OF CHRONIC MITRAL REGURGITATION

PRIMARY	SECONDARY (ANNULUS DILATATION)
• Mitral valve prolapse • Rheumatic fever (children) • Infective endocarditis • Atrial septal defect (Ostium primum) ◦ Ostium primum - cleft mitral valve	• Dilated cardiomyopathy ◦ Cause - Alcohol • Hypertrophic cardiomyopathy ◦ Systolic anterior movement of mitral valve → leakage of blood • Long standing Atrial fibrillation with left atrial enlargement

MECHANISM

00:14:32

- Leakage of blood from left ventricle (↑ pressure) to left atrium (↓ pressure)
 - Area of maximum velocity of blood (Narrowest part of jet) - **Vena contracta**



CRITERIA FOR SEVERE MITRAL REGURGITATION

- Vena contracta > 0.7 cm
- Regurgitation volume > 60 ml/beat
- Regurgitant fraction > 50%

CLINICAL FEATURES

- Simple lipids
- Exertional Fatigue
- B/L Ankle edema

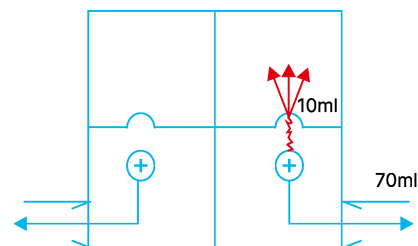
ON EXAMINATION

00:18:55

- Congestion in Jugular Veins (JVP ↑)
- BP - Normal /↓
- Palpable thrill at apex /Hyperdynamic apex beat of heart
 - In chronic mitral regurgitation
 - Left ventricle hypertrophy (due to volume overloading)
- Apex beat displacement
- Hepatomegaly
- B/L Pitting ankle edema
- Auscultatory findings



- S1 - Soft
- S2 - A2P2
 - P2 varying with respiration
 - Backflow of blood → ↓ blood to be pumped (to aorta) → Early closure of Aortic valve → ↑ gap between A2 & P2
 - Wide variable split S2
- S3 - heard (in Chronic mitral regurgitation)
- Pan systolic murmur in Apex, which radiates to axilla
 - Inability of leaflets to coapt → backflow /leakage of blood (to left atria)



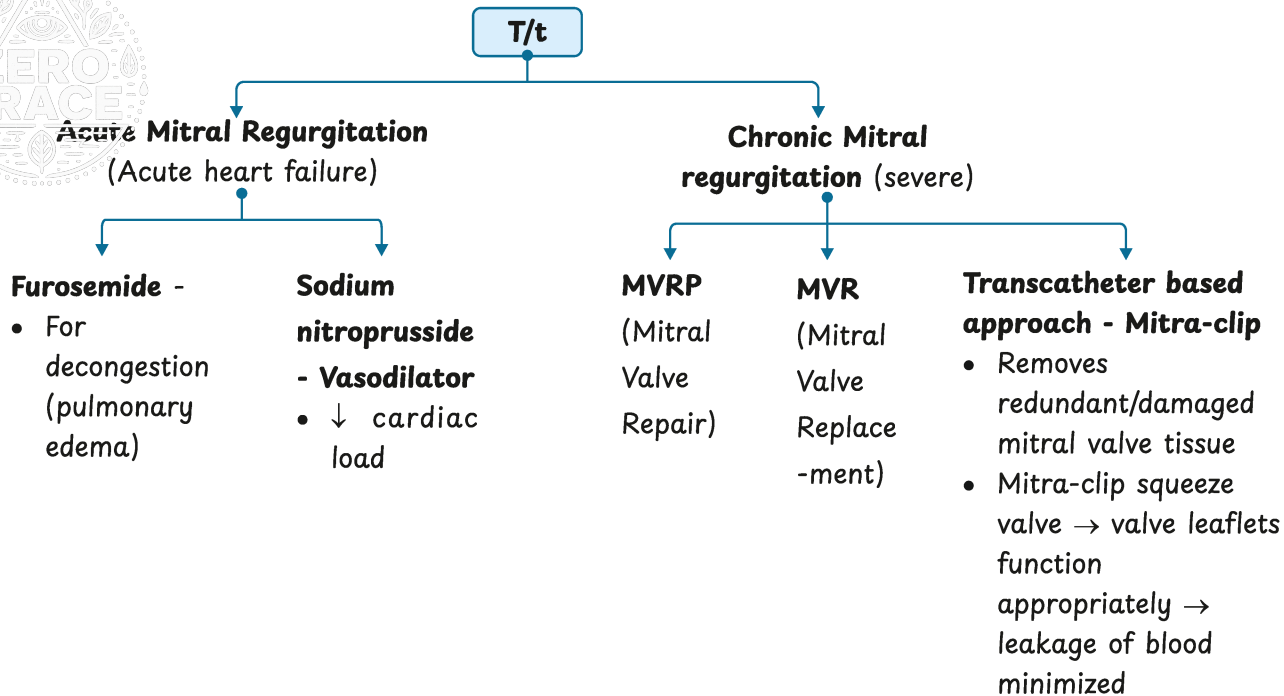
ON AUSCULTATION

• Chronic Mitral Regurgitation - ◦ Pansystolic murmur (Uniform intensity throughout systole) ◦ Flow murmur (mid-diastolic murmur)	
• Mitral valve prolapse - ◦ Mid-systolic click (stretching of chordae tendinae) ◦ Late systolic murmur (crescendo-decrescendo murmur)	
• Acute Mitral Regurgitation (after MI) - ◦ Early systolic murmur (papillary muscle dysfunction)	

WORK UP

00:32:45

ECG	CHEST X-RAY	TRANSTHORACIC ECHOCARDIOGRAPHY
• P-mitrale (Pulmonale) ◦ Notch in middle of P-wave (vertical height ↑) ◦ Duration of P-wave > 120 msec ◦ Seen in left atrial enlargement	• Double atrial shadow/contour • Cardio-thoracic ratio ↑ ◦ > 0.5 • Carinal angle ↑ ◦ > 60-70 degrees ◦ Normal - 60 - 70 degrees • Walking man sign	• IOC



- **Atrial fibrillation** (seen in cases of Mitral regurgitation)
 - Structural damage to left atria → Clot formation
 - Predisposes to embolization → Cerebral embolism → ↑ Risk of Stroke
 - Etiology -

RHEUMATIC FEVER INDUCED ATRIAL FIBRILLATION	MITRAL VALVE PROLAPSE INDUCED ATRIAL FIBRILLATION
T/t - Oral Anticoagulants (Warfarin) → ↓ risk of stroke	T/t - Novel oral Anticoagulants (Apixaban, Edoxaban)

Important Information

- Novel Anticoagulants C/I in -
 - Atrial fibrillation (due to Rheumatic heart disease)
 - Causes mitral stenosis & mitral regurgitation
 - Metallic prosthetic heart valve - Warfarin

MCQs

00:45:30

- Q. A 23-year-old male presents to the clinic complaining of fatigue, shortness of breath, & occasional palpitations. Echocardiography shows back-flow from left ventricle to left atrium during systole. Which of the following is not correct about this condition?
- Mid-diastolic murmur
 - Soft 1
 - Wide split S2
 - Carotid thrill

Ans: (d)



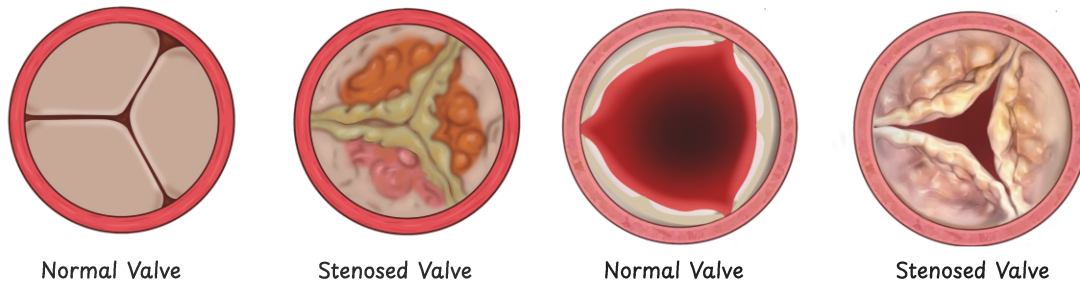
6. AORTIC STENOSIS



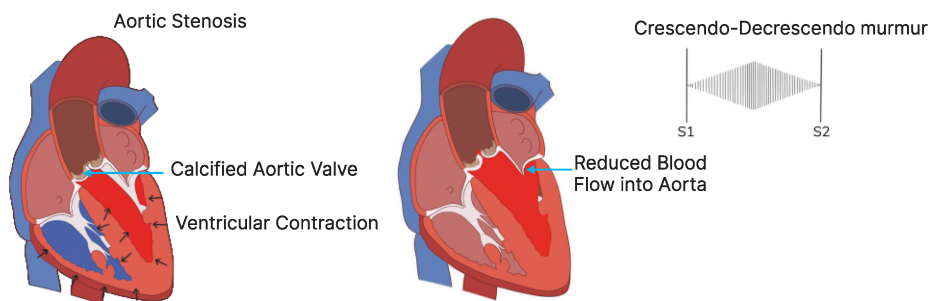
AORTIC STENOSIS

HEMODYNAMICS

00:00:48



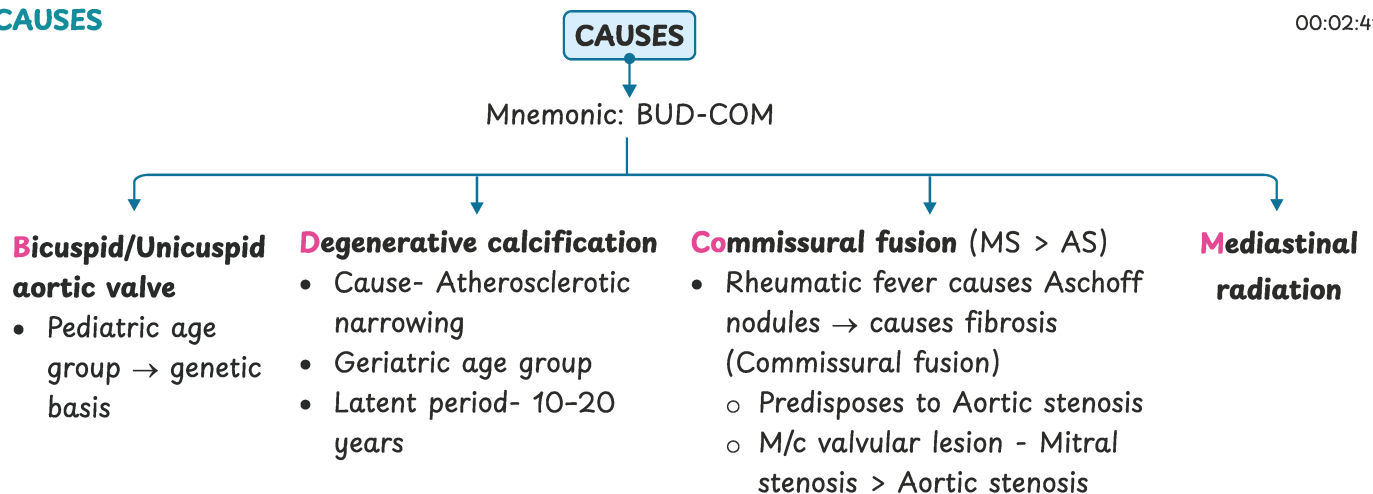
- Left ventricles have to work against resistance
 - Cause- Calcified aortic valves
 - Blood pumped is lesser
 - Blood pumped through narrow orifice - Ejection systolic murmur



- Seen in extreme age groups
 - Pediatric age group (infantile), Geriatric age group

CAUSES

00:02:41



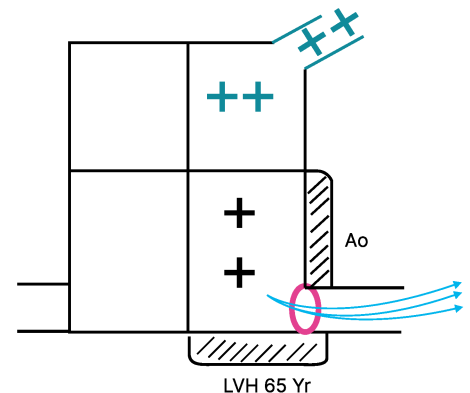
Important Information

- M/c congenital valvular lesion - Aortic stenosis
- M/c congenital heart disease in Turner syndrome - **Bicuspid aortic valve** > Coarctation of aorta
- Bicuspid aortic valve -
 - Genetic disorder, **autosomal dominant**
 - Gene - **Notch 1 gene**
- Heart valve least likely to be involved in Rheumatic etiology - **Pulmonic valve**
- NOTCH 3 gene → responsible for
 - Cerebral autosomal dominant arteriopathy with subcortical infarcts and leukoencephalopathy (CADASIL) → causing small vessel stroke

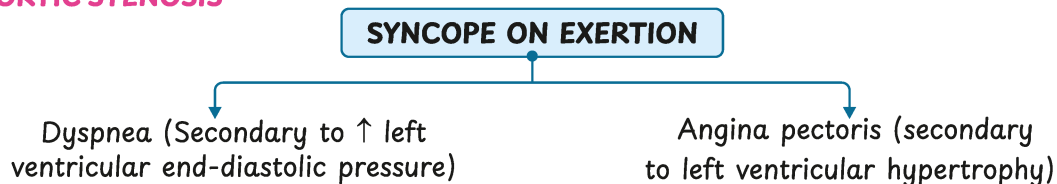
CLINICAL FEATURES

00:09:09

- Normal size of aortic valve orifice - $3-4 \text{ cm}^2$
- Size of aortic valve orifice (Severe aortic stenosis) - $< 1 \text{ cm}^2$ / $< 0.6 \text{ cm}^2 / \text{m}^2 \text{BSA}$ (Body Surface Area)
- Compensatory Left Ventricular Hypertrophy
 - LV generate more force & pressure (than aorta)
 - Transvalvular pressure gradient
→ Left Ventricle - Aorta > 40 mm Hg
 - Indicative of **Severe aortic stenosis**
- Ejection velocity > 4 m/sec (Severe aortic stenosis)
 - Normal Ejection velocity - 1 m/sec



TRIAD OF AORTIC STENOSIS



- In Left ventricular hypertrophy (LVH)
 - Blood supply - Normal, ↑ Demand
 - Narrowing of coronary arteries
 } Causes angina pectoris (chest pain)
- 1st Comorbidity (of Aortic stenosis) - Double/Triple vessel disease (causing angina pectoris)
- ↑ ventricular end - diastolic pressure → Congestion in atria & lung circulation → Pulmonary edema

EXAMINATION FINDINGS

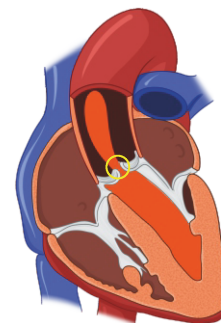
PULSE	• Pulsus parvus et tardus - Slow rise of pulse ◦ Delay in peak of pulse ◦ Cause - ↓ Volume of blood pumped from heart
BP	• Hypertension ◦ 2nd Comorbidity (of Aortic stenosis) ◦ Concomitant condition with Aortic Stenosis in geriatric patients



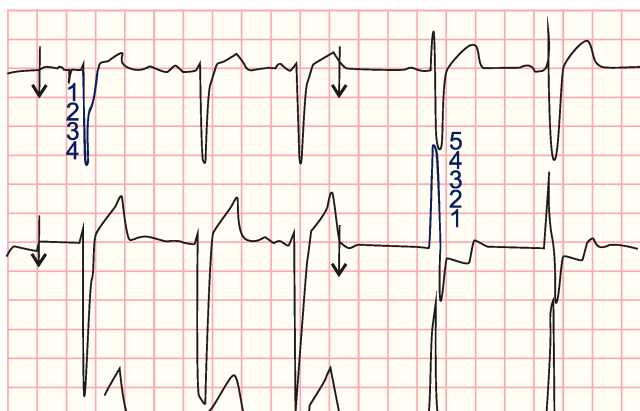
	- Displaced lateral to mid-clavicular line - Heaving in nature (forceful, sustained impulse) - Double apical impulse - Apical impulse felt on systole & diastole - Cause-Forceful left atrial contraction against less / non-compliant Left Ventricle (2^o to LVH) → cause 2nd impulse
CAROTID THRILL	- Thrill → Palpable murmur - Narrowed orifice → ↑ Ejection velocity (left ventricular hypertrophy) → loud noise produced → can be palpated
HEART SOUNDS	- S1 - Soft (severe), normal (mild) - S2 - A2P2 - Aortic valve calcified → feeble A2 component ↓ Intensity of S2 → Soft S2 Narrow split S2 (Gap between A2 & P2 < 30 millisecond) ↑ time required for exit of blood ↓ Gap between Aortic valve closure & pulmonic valve closure Single S2 - Aortic valve & pulmonic valve close at same time Reverse split S2 (P2A2) - Aortic valve closes after pulmonic valve closure - S4 - Atrial gallop rhythm - Cause - Left Atrium hypertrophy
MURMUR	Ejection systolic murmur /crescendo – decrescendo murmur - Increase and decrease in intensity of murmur - Valsalva maneuver/Patient made to stand - Pooling of blood peripherally → ↓ entry of blood to heart → Murmur becomes softer - Present at right upper sternal border
GALLAVARDIN PHENOMENON	- Harsh ejection systolic murmur → radiation to carotid arteries - Presence of ↑ pressure gradients → Vibrations of valve leaflets → radiates to axilla (mimics murmur of mitral regurgitation) - Gallavardin phenomenon

Important Information

- Bruit - Turbulence due to narrowing of arteries
 - Eg - Renal artery stenosis
- Dancing carotids -
 - Seen in Valvular aortic regurgitation



- LVH (Depth of SV1 + Height of RV6) > 35 mm



2. Chest X ray

- Cardio- thoracic ratio ↑ (> 0.5)
- Apex - Infero-lateral displacement

3. Transthoracic echocardiography

- IOC

VALVE SIZE	GRADE OF AS
1.5 - 2 cm ²	Mild AS
1.0-1.5 cm ²	Moderate AS
< 1.0 cm ²	Severe AS

4. Coronary Angiography

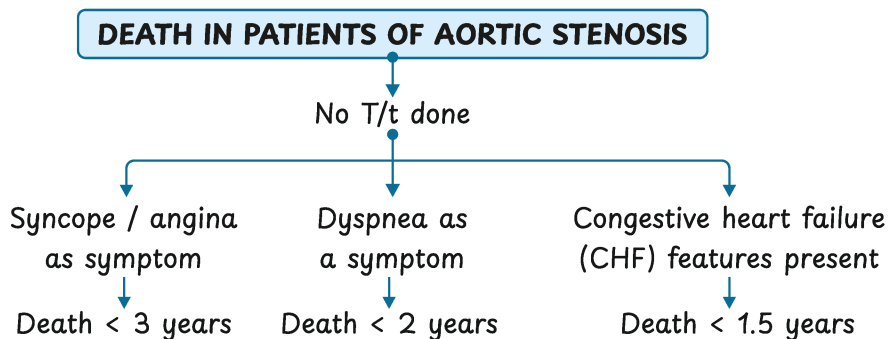
- To evaluate for obstruction in coronary arteries
- If patient not willing → **Coronary CT Angiography** can also be done
- Atherosclerotic plaques identified
 - Contributes to narrowing of coronary vessels

5. CT Chest/ Cardiac MRI

- Evaluates cardiac muscles & aortic valvular status (Senile degeneration/Calcific degeneration/fibrosis)

6. Treadmill test

- Absolute C/I - Severe aortic stenosis



- The latency period on onset is 10-20 years

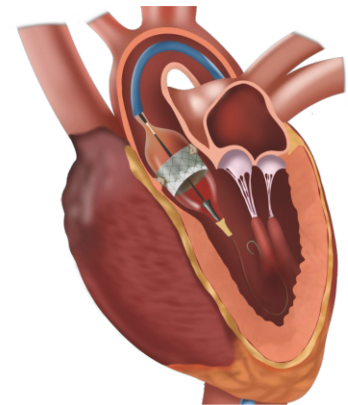
STAGES OF AORTIC STENOSIS

00:40:12

STAGE A	• Risk factors for valvular malfunction present
STAGE B	• Mild -moderate aortic stenosis
STAGE C	• Asymptomatic Severe aortic stenosis
STAGE D	• Symptomatic Severe aortic stenosis

T/t

00:44:19



HTN

- Careful titration of dose of ACE inhibitors

Statins

- Due to presence of atherosclerosis

No competitive sports

Avoid dehydration

Sublingual Nitroglycerine

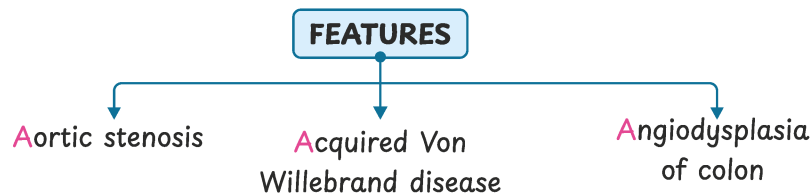
- For angina pectoris

TAVI

- Transcatheter Aortic Valve Implantation (TAVI)
- Over calcified valve → new valve deployed
- Indication -
 - Stage D aortic stenosis (Symptomatic)
 - $V_{max} > 4$ msec
 - Transvalvular pressure gradient (Left Ventricle - Aorta) > 40 mmHg
 - Stage C aortic stenosis - Asymptomatic with below risk factors
 - $V_{max} > 4$ msec
 - Transvalvular pressure gradient (Left Ventricle - Aorta) > 40 mmHg
 - Left ventricular ejection fraction $< 50\%$

HEYDE SYNDROME

00:46:55





Q. A 60 year old hypertensive female presents to the clinic with complaints of exertional dyspnea and angina episodes. A systolic murmur is heard over the right upper sternal border on cardiac auscultation. Echo shows transvalvular gradient of 50 mm Hg across aortic valve. Which of the following is not correct about this condition?

- a. Pulsus parvus et tardus
- b. Loud A2
- c. Carotid thrill
- d. Murmur decreases with Valsalva maneuver

Ans. (b)

Q. A 65-year-old male presented to the clinic due to shortness of breath on exertion, chest pain, and occasional fainting episodes. His blood pressure is elevated. Cardiac auscultation reveals a narrow split S2 and a harsh murmur. Which of the following is correct about character of this murmur?

- a. Ejection systolic murmur
- b. Holosystolic murmur
- c. Mid systolic click followed by systolic murmur
- d. Opening snap followed by mid-diastolic murmur

Ans. (a)



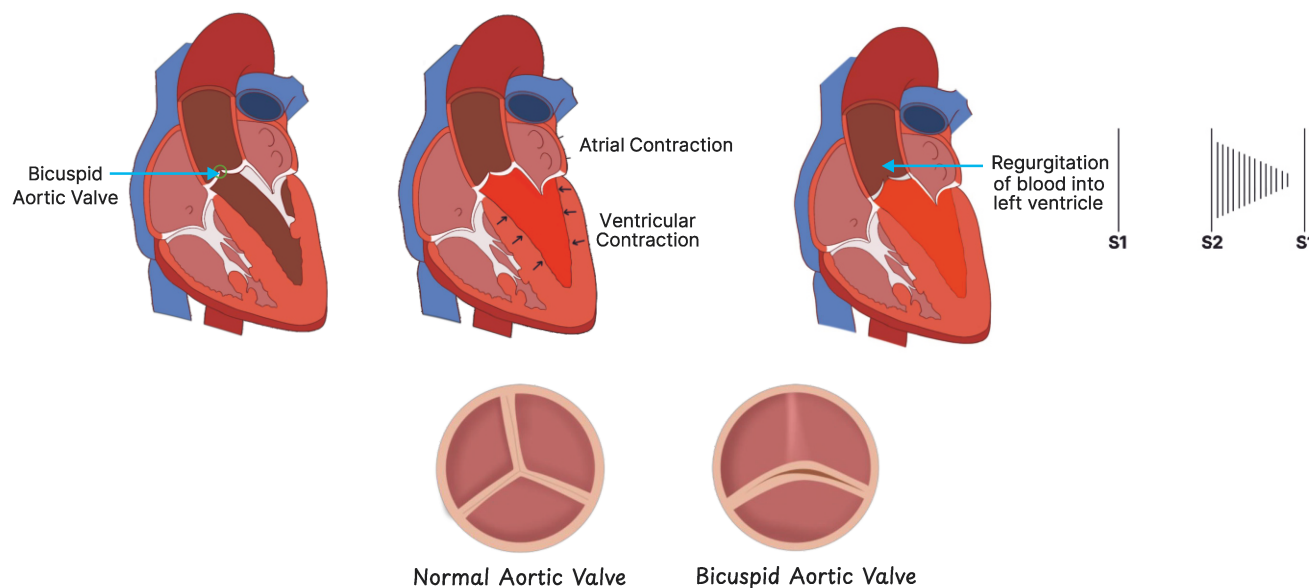
AORTIC REGURGITATION



AORTIC REGURGITATION

CAUSES

00:02:52



- **Bicuspid aortic valve**
 - Ventricular contraction → Blood pumps to aorta → substantial amount of blood leaks back into left ventricle (from aorta)
 - Acute - Left ventricular failure
 - Chronic - Left ventricular hypertrophy
 - Causes concomitant Aortic Stenosis and Aortic Regurgitation
- **Ankylosing spondylitis- HLA B27 +**
 - Causes early diastolic murmur (Blood leaks from aorta to left ventricle)
 - Severe aortic regurgitation → Blood strikes anterior leaflet of mitral valve → Causes Austin Flint murmur
- **Endocarditis**
 - Bacterial vegetations causing perforation on valve → blood leaks back to left ventricle (from aorta)
- **Aortic root dilatation -**
 - Seen in - Marfan's syndrome, Hypertension, Takayasu's arteritis
- **Rheumatic heart disease**
 - Causes cusp shortening → Blood leaks back to aorta
- **Myxomatous degeneration of aortic valve cusp**
- **Syphilis**



ECHOCARDIOGRAPHIC CRITERIA

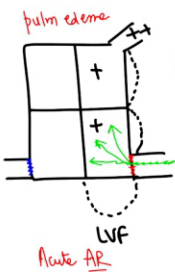
Severe aortic regurgitation-

- Regurgitant volume > 60 ml/ beat
- Regurgitant fraction > 50%
- Diastolic flow reversal of thoracic aorta
 - Head nodding (*De Musset's sign*)

Progressive aortic regurgitation -

- Left Ventricular Ejection Fraction < 50%
- Left Ventricular End Systolic Dimension > 50 mm
- Left Ventricular End Diastolic Dimension > 65 mm

PRESENTATION

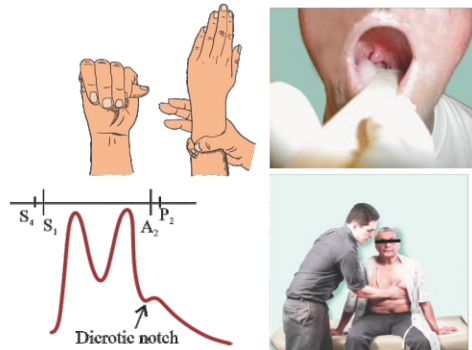
ACUTE PRESENTATION	CHRONIC PRESENTATION
	
• Substantial amount of blood leaking back into LV from aorta → Pressure rises in left ventricle & left atria → Left Ventricular and Left Atrial dilatation → Severe pulmonary edema • Seen in Infective endocarditis	• Minimal leakage of blood into LV from aorta → stretching of wall of LV → ↑ force of contraction ◦ <i>Frank Starling law</i> • Gives rise to fast rise & fast fall pulse - <i>Collapsing/water hammer</i> pulse • Eccentric hypertrophy of heart (LV dimension ↑) • Coronary blood flow - ↓ ◦ Diastolic BP - ↓ ◦ Causes Chest pain

CLINICAL FEATURES - CHRONIC AORTIC REGURGITATION

00:18:42

CLINICAL FEATURES

Latency period - 10- 15 years	• In Supine position - discomfort in chest ◦ Cause - Backflow of blood into LV	• Head nodding ◦ Cause - To-and-fro movement of head → abrupt distension of blood vessels	Shortness of breath / Dyspnea on exertion/ Paroxysmal Nocturnal Dyspnea/ Orthopnea	Angina	Ankle edema
-------------------------------	---	--	--	--------	-------------



MULLER SIGN	- Pulsation of uvula - Cause - Abrupt distension of blood vessels
DE-MUSSET SIGN	- Head nodding / bobbing - Cause - Abrupt distension & collapse of carotid arteries
THUMB SIGN / STEINBERT SIGN	- Ask patient to make a fist → projection of thumb - Indicates Marfan's syndrome - Arachnodactyly
WRIST SIGN/ WALKER MURDOCH SIGN	- Ask patient to encircle wrist using thumb & little finger → terminal phalanx of little finger overlaps thumb - Indicates Marfan's syndrome
MODIFIED SCHOBER TEST	- Extreme stiffness → not able to bend forward (to touch knees) - Indicates Ankylosing spondylitis - HLA B27 positive
CORRIGAN'S PULSE / WATER HAMMER PULSE / COLLAPSING PULSE	Evaluate radial artery (apply hand & apply pressure while encircling wrist - Obliterate radial artery) → raise arm of patient (continued pressure applied) → backflow of blood occurs → heart contracts more forcefully → Forceful pulse
TRAUBE SIGN	Pistol shot sounds heard over femoral artery on auscultation
DUROZIEZ SIGN	Systolic and diastolic bruit appreciated over femoral artery
QUINCKE'S SIGN	Pulsations in proximal part of nail bed
PULSE PRESSURE	↑ Systolic BP ↓ Diastolic BP- ↓ Aortic recoil - Pulse Pressure = Systolic BP - Diastolic BP, Increased - Abnormal BP finding - - Example: 160/0 mm Hg → Not actual BP finding → Korsakoff sounds to be evaluated till Muffled (Eg:- 160/60 mm Hg)



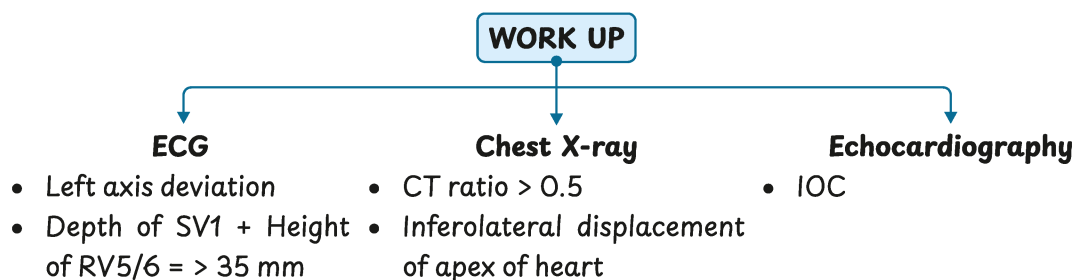
BLOOD PRESSURE (UL) VS LOWER LIMB (LL)	• Normal ◦ UL BP < 120/80 mm Hg ◦ LL BP < 140/80 mm Hg ◦ Difference < 20 mm Hg • Chronic Aortic Regurgitation ◦ UL BP - 160/60 mm Hg ◦ LL BP - 190/60 mm Hg ◦ Difference > 20 mm Hg → Hill sign
APEX BEAT	• Displaced, Heaving type
HEART SOUNDS	• Soft S2 • Ejection click present ◦ If bicuspid aortic valve is present • Early diastolic murmur (Decrescendo/ Seagull murmur) ◦ Murmur of aortic regurgitation • Mid diastolic murmur - Due to vibrations in mitral valve ◦ Murmur of severe aortic regurgitation → Austin Flint murmur
GERHARDT'S SIGN	• Pulsatile spleen
ROSENBACH SIGN	• Pulsatile liver • Seen in - ◦ Tricuspid regurgitation, Tricuspid stenosis, Aortic regurgitation
LANDOLFI SIGN	• Pulsatile pupil

Important Information

- Abnormal BP finding (Eg - BP - 160/0)
 - Seen in -
 - AV fistula
 - Aortic regurgitation
 - Arteriovenous malformation
- In coarctation of aorta - ↑ UL BP, ↓ LL BP
- Traube's space - Spleen enlargement
- Rosenbach sign - Grave's disease → sympathomimetic stimulation → eyelid tremor

WORK UP

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ACUTE AR	CHRONIC AR
• Acute heart failure ○ I/V furosemide ○ Vasodilator - Sodium Nitroprusside ○ Surgery for valve replacement (within 24 hours) → To resolve pulmonary edema ○ C/I → IABP - Intra- Aortic Balloon pump → β - blockers	• ACE inhibitors - Maintain SBP < 130 mm Hg • S/L Nitroglycerin- For Angina • β - blockers • Aortic valve replacement - For severe symptomatic Aortic regurgitation